

Project Overview

Project Overview

1.1 Introduction

With the rapid growth of digital media, educational institutions increasingly use social media platforms to promote their campuses, communicate with prospective students, publish research achievements, announce rankings, and engage with the public. Among these platforms, YouTube serves as one of the richest sources of publicly available educational content, containing thousands of videos related to university infrastructure, student life, controversies, research activities, rankings, and institutional branding.

However, this vast amount of information is highly unstructured. Videos are uploaded by official university channels, coaching institutes, educational influencers, news agencies, students, alumni, and the general public. Consequently, the collected data often contains irrelevant videos, duplicate records, incomplete descriptions, inconsistent formatting, and noisy textual information that must be processed before any meaningful analysis can be performed.

This project aims to build a complete data preprocessing and analytical pipeline capable of transforming raw YouTube metadata into a structured dataset suitable for Natural Language Processing (NLP), sentiment analysis, topic modelling, and comparative educational research.

Unlike many studies that rely on manually curated datasets, this project develops an automated preprocessing framework that can be reused across multiple educational themes. The framework ensures that only relevant, high-quality, and standardized textual data are used for downstream analysis.

1.2 Project Objective

The primary objective of this project is to collect, preprocess, and analyze YouTube videos related to Indian universities across multiple educational themes.

The project focuses on transforming noisy raw YouTube metadata into a clean and structured dataset that can later be used for:

- Exploratory Data Analysis (EDA)
- Sentiment Analysis using VADER
- TF-IDF Keyword Analysis
- Topic Modelling using Latent Dirichlet Allocation (LDA)

- Hashtag Analysis
- Network Analysis
- Comparative Analysis among different educational themes

The preprocessing pipeline has been designed to ensure data consistency, reproducibility, and scalability across different datasets.

1.3 Research Objectives

The project has the following research objectives:

1. Collect publicly available YouTube videos related to Indian universities.
 2. Organize videos into multiple educational categories.
 3. Remove irrelevant videos using an automated relevance scoring algorithm.
 4. Handle missing and inconsistent metadata.
 5. Clean textual information while preserving semantic meaning.
 6. Engineer additional numerical and textual features useful for analytical tasks.
 7. Validate the final dataset before applying NLP techniques.
 8. Generate structured datasets suitable for educational social media analysis.
-

1.4 Dataset Categories

The collected dataset is divided into multiple thematic categories.

Infrastructure

Videos related to:

- Campus Tours
- Hostel Facilities
- Classrooms
- Laboratories
- Libraries
- Sports Facilities
- Mess and Food Courts
- Campus Life
- Student Experience

Controversies

Videos discussing:

- Student Protests
- Institutional Scandals
- Ragging Incidents
- Fraud Cases
- Fake News
- Legal Issues
- Public Complaints
- Administrative Actions

Faculty Research

Videos related to:

- Faculty Publications
- Research Projects
- Academic Conferences
- Journal Publications
- Research Ethics
- Plagiarism
- University Research Achievements

Rankings

Videos discussing:

- NIRF Rankings
 - NAAC Accreditation
 - Global Rankings
 - Institutional Reputation
 - Ranking Comparisons
-

1.5 Dataset Source

The dataset was collected using the **YouTube Data API v3**.

Only publicly available video metadata was collected.

The dataset contains information such as:

- Video ID
- University Name
- Video Title
- Video Description
- Channel Name
- Upload Date
- Upload Year
- Number of Views
- Number of Likes
- Number of Comments
- Video Duration
- Language
- Video URL

Additional metadata such as category, topic, and search query were also collected where applicable.

No private user information or personal data was collected.

1.6 Technologies Used

The preprocessing pipeline was implemented entirely in Python.

The following libraries were used:

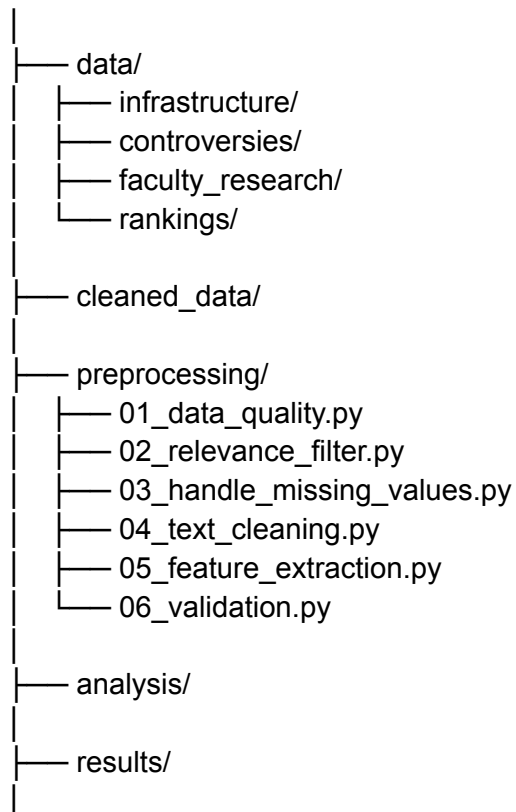
Library	Purpose
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pandas	Data manipulation and preprocessing
NumPy	Numerical operations
Regular Expressions (re)	Text cleaning
html	HTML entity decoding
RapidFuzz	Fuzzy similarity matching
math	Logarithmic transformations
datetime	Video age calculation
os	File handling

1.7 Project Folder Structure

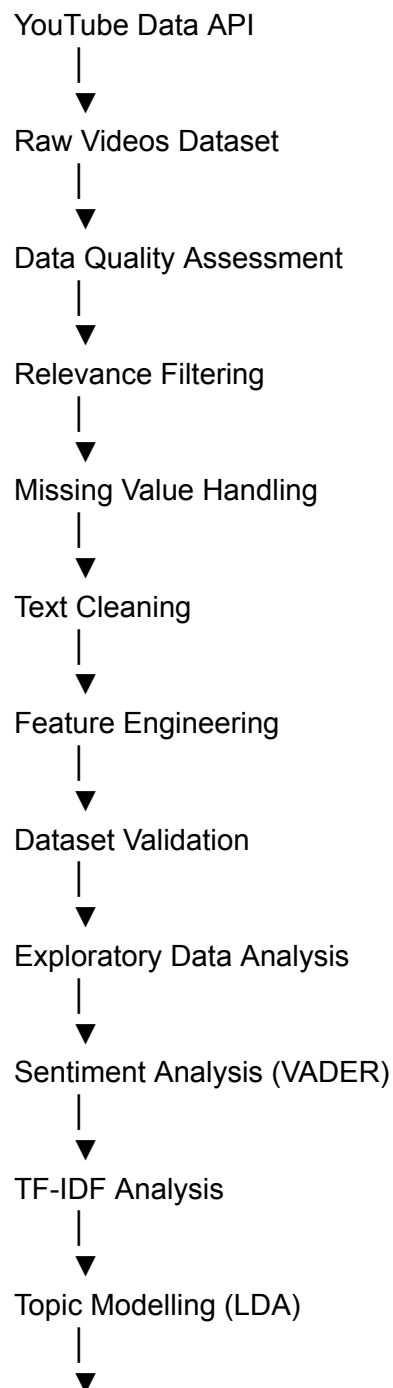
The project follows a modular folder structure to ensure maintainability and reproducibility.

youtube_university_analysis/



1.8 Overall Workflow

The complete workflow of the project is illustrated below.



Hashtag Analysis



Network Analysis



Comparative Educational Insights

1.9 Importance of Preprocessing

Raw YouTube metadata is inherently noisy and inconsistent. Videos collected using search queries may include unrelated content, incomplete descriptions, inconsistent formatting, HTML entities, URLs, duplicate records, and missing metadata. Applying analytical techniques directly to such data can produce misleading results.

Therefore, a structured preprocessing pipeline was developed to ensure that only relevant, clean, and validated data are used for subsequent analysis. The preprocessing stage standardizes textual information, removes unnecessary noise, engineers meaningful features, and verifies dataset integrity before any NLP or statistical methods are applied. This process improves the reliability, reproducibility, and overall quality of the research outcomes.

Data Collection Methodology

Data Collection Methodology

2.1 Introduction

The quality of any data-driven research depends heavily on the quality of the collected dataset. Since this project focuses on analyzing public perceptions and narratives related to Indian universities, the first stage involved collecting publicly available YouTube videos from multiple educational domains.

Instead of manually collecting videos, an automated data collection pipeline was developed using the **YouTube Data API v3**. This approach enabled systematic retrieval of metadata from thousands of videos while ensuring consistency, scalability, and reproducibility. The collected data forms the foundation of the preprocessing and analytical stages described in later chapters.

2.2 Why YouTube?

YouTube was selected as the primary data source because it is one of the largest repositories of educational, informational, and opinion-based content. Universities, students, alumni, coaching institutes, educational influencers, news agencies, and independent creators actively publish videos discussing various aspects of higher education.

Compared with other social media platforms, YouTube provides:

- Large volumes of publicly accessible educational content.
- Rich textual metadata including titles and descriptions.
- Quantitative engagement metrics such as views, likes, and comments.
- Content covering both official institutional perspectives and public opinions.
- API support for structured and automated data collection.

These characteristics make YouTube an appropriate platform for studying educational narratives at scale.

2.3 Data Collection Tool

Data collection was performed using the **YouTube Data API v3**, which provides programmatic access to publicly available video metadata.

The API was used to retrieve structured information for videos matching predefined search queries. Only publicly available metadata was collected. No user-identifiable information or private content was accessed.

Python scripts were developed to automate the collection process, ensuring that videos from multiple universities and multiple themes were gathered consistently.

2.4 University Selection

The study focuses on Indian universities representing different institutional categories and geographic regions.

The dataset includes universities from:

- Indian Institutes of Technology (IITs)
- National Institutes of Technology (NITs)
- Indian Institutes of Information Technology (IIITs)
- Central Universities
- State Universities
- Private Universities
- Deemed Universities

The selected institutions represent diverse academic environments, allowing comparative analysis across different types of higher education institutions.

Approximately **90+ universities** were included in the final dataset.

2.5 Dataset Categories

Instead of collecting videos without any organization, the dataset was divided into thematic categories. This categorization enables focused analysis and comparison across different aspects of university-related discussions.

The following categories were created:

Infrastructure

Videos related to physical campus facilities and student life, including:

- Campus tours
 - Hostel tours
 - Classrooms
 - Laboratories
 - Libraries
 - Sports facilities
 - Mess and cafeteria
 - Campus reviews
 - Student experience
-

Controversies

Videos discussing institutional issues, including:

- Student protests
 - Administrative disputes
 - Ragging incidents
 - Scams
 - Fraud allegations
 - Legal proceedings
 - Viral incidents
 - Public complaints
-

Faculty Research

Videos focusing on academic activities, including:

- Research publications
- Faculty achievements
- Conferences
- Journals
- Research ethics
- Plagiarism
- Research funding
- Academic innovation

Rankings

Videos related to institutional rankings and reputation, including:

- NIRF Rankings
 - NAAC Accreditation
 - QS Rankings
 - University comparisons
 - Reputation analysis
-

2.6 Search Strategy

A keyword-based search strategy was adopted to maximize coverage while maintaining thematic relevance.

Each university was combined with category-specific search terms.

Examples include:

Infrastructure

IIT Delhi campus tour
VIT hostel review
NIT Trichy campus life
SRM classroom
LPU infrastructure

Controversies

JNU protest
Anna University scam
Lucknow University controversy
University ragging
Student protest university

Faculty Research

University research
Faculty publication

Research paper
Professor achievement
Research ethics

Rankings

NIRF ranking
NAAC accreditation
Best engineering colleges
University ranking India

This systematic search strategy ensured that videos represented different educational themes while minimizing irrelevant retrieval.

2.7 Metadata Collected

For every retrieved video, the following metadata were collected.

Attribute	Description
Video ID	Unique YouTube identifier
University Name	Associated university
Category	Dataset category
Topic	Theme of the search
Search Query	Query used to retrieve the video
Title	Video title
Description	Video description
Channel Name	Publishing channel
Upload Date	Publication date
Upload Year	Extracted publication year
Views	Total view count

Likes	Total likes
Comment Count	Total comments
Duration	Video duration
Language	Detected language
Video URL	Public YouTube URL

This metadata provides both textual and numerical information required for subsequent preprocessing and analysis.

2.8 Dataset Organization

To improve maintainability, datasets were organized into separate folders according to category.

```
data/
|
|— infrastructure/
|   videos.csv
|
|— controversies/
|   videos.csv
|
|— faculty_research/
|   videos.csv
|
|— rankings/
|   videos.csv
```

Each folder contains the raw dataset collected directly from the YouTube Data API.

This modular organization simplifies preprocessing, analysis, and future dataset expansion.

2.9 Data Collection Challenges

Several challenges were encountered during data collection:

- Many search results were unrelated to the intended university or topic.
- Different universities often share similar abbreviations (e.g., "DU", "IIT", "BIT"), leading to ambiguous search results.
- Some videos lacked descriptions.
- Videos uploaded by unrelated channels occasionally matched search keywords.
- Duplicate videos appeared across multiple search queries.
- Public engagement metrics varied significantly between videos.

These challenges motivated the development of a dedicated preprocessing pipeline to improve dataset quality.

2.10 Need for Preprocessing

The collected dataset contained several issues that prevented direct analysis, including:

- Irrelevant videos.
- Missing textual information.
- Inconsistent formatting.
- HTML entities.
- URLs embedded in descriptions.
- Duplicate records.
- Unstructured textual metadata.
- Noisy content.

Consequently, a multi-stage preprocessing pipeline was designed to transform the raw dataset into a clean, standardized, and analysis-ready dataset.

2.11 Transition to Preprocessing

Following data collection, each dataset underwent a structured preprocessing workflow consisting of six stages:

1. Data Quality Assessment
2. Relevance Filtering
3. Missing Value Handling
4. Text Cleaning
5. Feature Engineering
6. Dataset Validation

Each stage addresses a specific aspect of data quality and prepares the dataset for downstream analyses such as Exploratory Data Analysis (EDA), sentiment analysis, TF-IDF, topic modelling, hashtag analysis, and network analysis.

Preprocessing Methodology Overview

Preprocessing Methodology Overview

3.1 Introduction

Data collected directly from the YouTube Data API cannot be used immediately for analytical tasks because it contains several forms of noise and inconsistency. Videos returned by keyword-based searches frequently include unrelated content, incomplete metadata, inconsistent formatting, duplicated information, and textual artifacts such as HTML entities, URLs, and irregular spacing.

If such data are used directly for Natural Language Processing (NLP), statistical analysis, or visualization, the resulting insights may be inaccurate or misleading. Therefore, a systematic preprocessing pipeline was developed to improve the quality, consistency, and usability of the collected dataset.

The preprocessing methodology adopted in this project follows a sequential workflow where the output of one stage becomes the input of the next stage. Each stage addresses a specific data quality issue while preserving information required for later analyses.

3.2 Objectives of Preprocessing

The preprocessing stage was designed with the following objectives:

- Evaluate the quality of the collected dataset.
 - Remove irrelevant videos retrieved through keyword-based searches.
 - Handle missing and incomplete metadata.
 - Standardize textual information.
 - Preserve meaningful textual features such as hashtags and emojis.
 - Create additional numerical and textual features useful for analysis.
 - Validate the processed dataset before applying NLP techniques.
 - Produce reproducible datasets suitable for research.
-

3.3 Why Preprocessing is Necessary

The raw YouTube datasets contained multiple issues that could negatively affect analytical performance.

Examples include:

Irrelevant Videos

Searching for university names sometimes retrieved unrelated videos.

Example:

University XYZ Movie Trailer

Although the university name appeared in the title, the video had no educational relevance.

Missing Descriptions

Many YouTube Shorts and promotional videos contained no descriptions.

Example:

Title:

IIT Delhi Hostel Tour

Description:

(empty)

Such videos still provide valuable information through their titles and therefore should not be discarded.

HTML Entities

Some titles contained encoded HTML symbols.

Example:

Hostel & Campus Tour

which should become

URLs

Descriptions frequently contained promotional links.

Example:

Visit:

<https://www.youtube.com/....>

Admission Form:

www.college.com

These URLs contribute no semantic information for NLP and were therefore removed.

Duplicate Information

Videos often appeared multiple times because different search queries retrieved the same YouTube video.

Duplicate records increase dataset size without providing additional information.

Inconsistent Text Formatting

Examples included:

- Multiple spaces
- New lines
- Tabs
- Repeated punctuation
- HTML formatting

Standardization improves tokenization and downstream NLP performance.

3.4 Preprocessing Philosophy

Rather than aggressively removing information, this project follows a **preservation-first approach**.

The objective is to remove only information that introduces noise while retaining information that may contribute to later analyses.

The following principles guided preprocessing:

Preserve Meaning

Text should remain readable after cleaning.

Preserve Sentiment

Emojis and punctuation often indicate emotion.

Example:

Amazing Campus 🥰🔥

Removing emojis would reduce sentiment information available to VADER.

Preserve Social Media Context

Hashtags were retained because they represent themes.

Example

#IITDelhi
#CampusTour

These hashtags will later be used for hashtag frequency analysis.

Preserve Mentions

Mentions such as

@iitdelhi

may later be used for network analysis.

Preserve Numbers

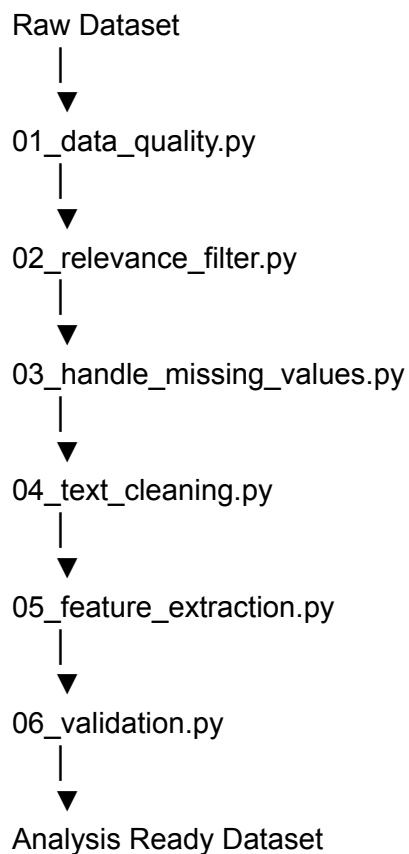
Numbers often represent:

- Rankings
- Years
- Fees
- Placement statistics

Therefore they were retained.

3.5 Overall Preprocessing Pipeline

The preprocessing pipeline consists of six independent scripts.



Each script performs a single responsibility, making the pipeline modular, reusable, and easy to maintain.

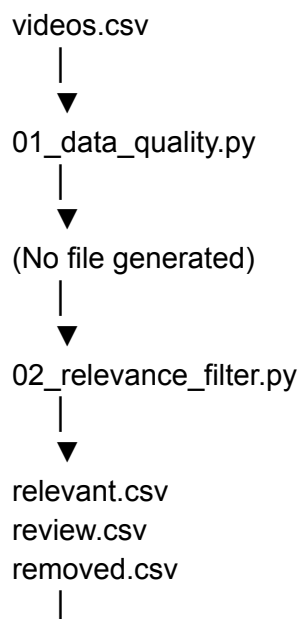
3.6 Preprocessing Scripts

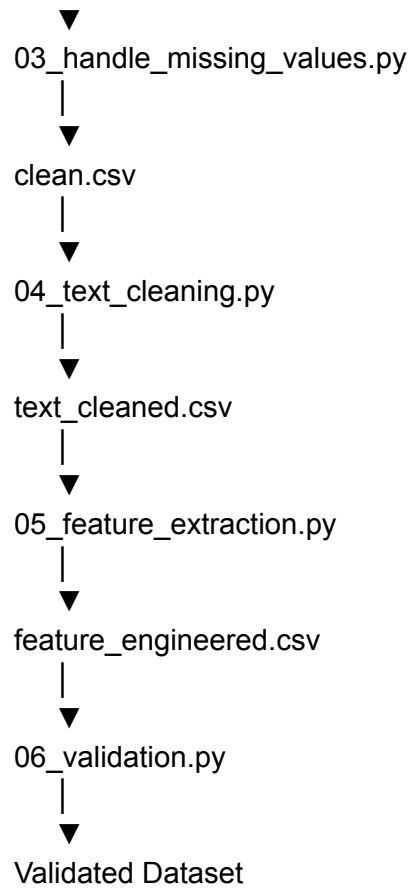
The preprocessing pipeline consists of the following modules.

Script	Purpose
01_data_quality.py	Evaluate dataset quality
02_relevance_filter.py	Remove irrelevant videos
03_handle_missing_values.py	Handle incomplete metadata
04_text_cleaning.py	Clean textual information
05_feature_extraction.py	Generate analytical features
06_validation.py	Verify processed dataset

3.7 Input and Output Flow

The output of each preprocessing stage becomes the input of the next stage.





3.8 Categories Processed

Each preprocessing script operates independently on four dataset categories.

- Infrastructure
- Controversies
- Faculty Research
- Rankings

The same preprocessing methodology is applied to each category to ensure consistency and enable meaningful comparative analysis.

3.9 Design Advantages

The proposed preprocessing pipeline provides several advantages:

- Modular implementation
 - Easy maintenance
 - Reusability across datasets
 - Consistent preprocessing
 - Reproducible workflow
 - High-quality analytical datasets
 - Suitable for NLP tasks
 - Suitable for statistical analysis
 - Easy integration with future datasets
-

3.10 Transition to Individual Scripts

Having described the overall preprocessing methodology, the following chapters explain each preprocessing script in detail. Every script is documented with its objective, theoretical background, implementation strategy, input and output files, generated results, and contribution to the overall preprocessing pipeline.

01_data_quality.py

Purpose

The objective of this script was to examine the quality of the collected raw YouTube datasets before preprocessing. This step helps identify missing values, duplicate records, data types, and other quality issues.

Input

data/infrastructure/videos.csv
data/controversies/videos.csv
data/faculty_research/videos.csv

Operations Performed

- Loaded dataset
 - Checked dataset dimensions
 - Displayed column names
 - Examined data types
 - Counted missing values
 - Identified duplicate rows
 - Checked duplicate Video IDs
 - Counted unique universities
 - Counted unique YouTube channels
 - Generated descriptive statistics
 - Calculated memory usage
 - Displayed sample records
-

Output (Infrastructure)

Dataset Shape

Rows : 7531

Columns : 16

Missing Values

Description : 2371

Duplicate Rows

0

Duplicate Video IDs

0

Unique Universities

93

Unique Channels

3668

Memory Usage

6.33 MB

Observation

- No duplicate records were found.
- A large number of videos contained missing descriptions.
- Dataset structure was consistent.

Output (Faculty Research)

Dataset Shape

Rows : 6448

Columns : 16

Missing Descriptions

1360

Duplicate Rows

0

Duplicate Video IDs

0

Unique Universities

92

Unique Channels

3429

Memory Usage

6.69 MB

Observation

Faculty Research videos had fewer missing descriptions than Infrastructure and showed no duplicate records.

02_relevance_filter.py

Purpose

Remove irrelevant YouTube videos collected during API scraping.

Input

*_clean.csv

Processing

- University name matching
- Alias matching
- Category keyword matching
- Search query similarity
- Negative keyword penalty
- Relevance score generation

Output (Infrastructure)

Total Videos : 7531
Relevant : 1947
Review : 3618
Removed : 1966

Average Score : 7.97
Highest Score : 54
Lowest Score : 0

Files Generated

infrastructure_relevant.csv
infrastructure_review.csv
infrastructure_removed.csv

Observation

Approximately **26%** of the collected videos were classified as highly relevant, while about **26%** were removed due to low relevance scores. The remaining videos were marked for manual review.

Output (Controversies)

Total Videos : 4064
Relevant : 1633
Review : 1437
Removed : 994

Observation

Controversy-related searches produced a relatively higher proportion of relevant videos because search terms were more specific.

Output (Faculty Research)

Total Videos : 6448
Relevant : 2457

Review : 2324
Removed : 1667

Observation

Faculty Research achieved the highest number of relevant videos among the processed categories.

03_handle_missing_values.py

Purpose

Handle incomplete metadata.

Input

videos.csv

Processing

- Filled missing descriptions with empty strings.
 - Filled missing language values.
 - Filled missing channel names.
 - Replaced missing numerical values with zero.
 - Removed rows with missing titles.
-

Output

Infrastructure

Missing descriptions filled : 2371

Missing after cleaning

Description : 0

Title : 0

Views : 0

Likes : 0

Comments : 0

Generated File

infrastructure_clean.csv

Observation

All important columns became complete while preserving rows with empty descriptions.

04_text_cleaning.py

Purpose

Clean textual metadata for NLP.

Processing

- HTML decoding
 - URL removal
 - Email removal
 - HTML tag removal
 - Extra space removal
 - Newline removal
 - Combined title + description
-

Output

Infrastructure

Rows : 1947

New Columns

clean_title
clean_description
combined_text

Generated File

infrastructure_text_cleaned.csv

Observation

The text became standardized while preserving emojis, hashtags, mentions, and numerical values for downstream sentiment and hashtag analysis.

05_feature_extraction.py

Purpose

Generate numerical and textual features for analytical tasks.

Features Created

title_word_count
title_char_count
description_word_count
description_char_count
combined_word_count
combined_char_count
emoji_count
hashtag_count
mention_count
exclamation_count
question_count
engagement_rate
log_views
log_likes
log_comments
video_age_days

Output (Infrastructure)

Rows : 1947

Columns : 34

Average Words : 27.33

Average Characters : 173.57

Average Emojis : 0.50

Average Hashtags : 2.07

Average Engagement Rate : 0.025840

Generated File

infrastructure_feature_engineered.csv

06_validation.py

Purpose

Validate the final processed datasets.

Validation Checks

- Missing values
 - Duplicate rows
 - Duplicate Video IDs
 - Required columns
 - Engineered columns
 - Negative values
 - Empty combined text
-

Output (Infrastructure)

Missing Values : 0

Duplicate Rows : 0

Duplicate Video IDs : 0

Negative Values : 0

Required Columns : Present

Engineered Columns : Present

Validation Status

SUCCESS

Observation

The Infrastructure dataset successfully passed all validation checks and was declared ready for analysis.
